

## Tutrugbu Partial ATR (Dis)harmony in Search

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Search, identifying the presence of a given configuration, can indirectly check if a string-peripheral segment is of a particular type: when, say, every high vowel from itself looks for the first vowel to its left, only the string-first high vowel, if there exists one, cannot find a preceding vowel. With this property, an analysis is proposed for the otherwise systematic ATR harmony between prefixes and a stem in Tutrugbu. (The proposed analysis then incorporates an addendum on the round harmony among some prefixes in the same language.)

keywords: Tutrugbu; harmony; advanced tongue root (ATR); round (rounded, rounding, labial); Search

The peculiarity that the string-first vowel can have a say on the ATR harmony in Tutrugbu (a Ghana-Togo Mountain language of Niger-Congo [Heine 2017: 277]) is examined from the perspective of (substance-free) Search (Mailhot and Reiss 2007, Dabbous et al. 2022). In the following, the description of this framework will be interspersed with the illustration of the phenomenon in question.

The plain Tutrugbu ATR harmony is shown in (1): the ATR value of the (alternating) prefix vowel(s) is determined by that of the first stem vowel (Essegbey 2009: 39, Essegbey 2019: 32).<sup>1</sup>

(1) simplex ATR harmony (McCollum et al. 2020: 219, 220; tones omitted)<sup>2</sup>

- |    |                      |    |                                |
|----|----------------------|----|--------------------------------|
| a. | ɔ-da ‘CL3-copper’    | b. | ka-tɪ-ba-ba ‘CL7-NEG-FUT-come’ |
|    | o-pete ‘CL3-vulture’ |    | ke-ti-be-ʃe ‘CL7-NEG-FUT-grow’ |

When the respective two items in (1a) and (1b) are contrasted with each other, it is sure that the prefixes alternate. Since the first stem vowel is –ATR for the upper item in both columns, the prefix vowels are realized as –ATR. The +ATR realization happens for the same reason in the lower part. For these alternating prefix vowels, I will assume that they are underspecified for the feature ATR, though specified for the features Syllabic, High, Low, Back, and Round: I ({+syllabic, +high, –low, –back, –round}) for i~ɪ; U ({+syllabic, +high, –low, +back, +round}) for u~ʊ; O ({+syllabic, –high, –low, +back, +round}) for o~ɔ. For the pairing e~a, in addition to the aforementioned non-specification, Low and Back are not specified, either. So, the capital

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<sup>1</sup> The emphasis on the first stem vowel is due to the existence of disharmonic stems like hɔ̃litsa ‘chameleon’ (Essegbey 2019: 35).

<sup>2</sup> On the surface, ɪ and ʊ are realized as ɛ and ɔ, respectively (McCollum et al. 2020: 218, McCollum and Essegbey 2020: 3, 27).

digraph  $\mathcal{A}E$  for  $e \sim a$  means the set  $\{+\text{syllabic}, -\text{high}, -\text{round}\}$ .<sup>3</sup>

The ATR agreement is then formalized with a Search-based rule below:

(2) ATR agreement

Search INR:  $[+\text{syllabic}]$ ; DIR: right; TRM:  $[+\text{syllabic}, \alpha\text{ATR}]$

Change INR  $\sqcup \{\alpha\text{ATR}\}$

‘Unify a vowel with the ATR value of the first ATR-specified vowel to its right.’

Search identifies if a given configuration is present. It consists of three parameters: i) the initiators (INRs) are *all* the segments whose feature bundles are supersets of the set which contains what is specified; ii) the direction (DIR) instructs Search to proceed either leftward or rightward from an INR; iii) the terminator (TRM) is the *first* segment whose feature bundle is a superset of the set which contains what is specified to the designated side of an INR. This operation of seeking stops when either the TRM is found or one of the two end domains is reached. In (2), every vowel is asked to look for the first vowel with an ATR value to its right.<sup>4</sup> If not specified otherwise, Change only happens to those INRs which each successfully found their corresponding TRM. And since what is specified in Change is not set containment, the pair of braces ( $\{ \}$ ) is used; as for the (familiar) pair of brackets ( $[ ]$ ), it is reserved for super-/sub-set (Bale et al. 2014: 243). In (2), an INR is unified with the ATR value of the found vowel from Search. Unification, as defined in Bale et al. (2014: 245), is like union. But, when an INR contains a valued feature which is opposite to what is to be unified with, the INR remains unchanged after the application of unification. For instance, when a  $-\text{ATR}$  vowel is unified with  $+\text{ATR}$ , the vowel is still  $-\text{ATR}$  afterwards. As for a  $+\text{ATR}$  or ATR-underspecified vowel, the unification with  $+\text{ATR}$  results in the vowel with  $+\text{ATR}$  specification.

The straightforwardness of the ATR agreement, however, is complicated: if both i) the first vowel of a given string is a high vowel, and ii) there exists a prefixal non-high vowel, it is  $-\text{ATR}$ -exponents that appear (McCollum et al. 2020: 220, 221). The first two examples below all have  $+\text{ATR}$  vowel in the stem, but, instead of the predicted  $+\text{ATR}$  realization, it is  $-\text{ATR}$  realization that shows up.

(3) disharmony under the specified conditions (McCollum et al. 2020: 219, 220, 221)

a.  $\text{bo-ba-fe}$  ‘1PL-FUT-grow’ (cf.  $\text{bu-ti-fe}$  ‘1PL-NEG-grow’)

b.  $\text{i-ti-ka-a-ba-ba-wu}$  ‘1SG-NEG-PFV-PROG-VENTIVE-VENTIVE-climb’

c.  $\text{bo-ba-di-wu}$  ‘1PL-FUT-ITIVE-climb’ (cf.  $\text{i-di-to}$  ‘1SG-ITIVE-cook’)

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<sup>3</sup>  $\varepsilon$  does not occur in “prefixes” (McCollum and Essegbey 2020: 3) / “affixes” (McCollum et al. 2020: 218).

<sup>4</sup> Following Bale and Reiss (2018: 451 vs. 454-455), I practice the identity use of Greek letter variables, and my understanding is that there are at least two occurrences of a given variable. Moreover, I assume that Search and Change must each contain at least one such occurrence for a specific variable used. I take these two moves, for now, as my working hypothesis.

(Notice that the comparison at the end of (3a, c) tells us that the relevant prefixes do alternate.) The disharmony, nonetheless, is partial. When the aforementioned conditions are met, but, there remain prefixal high vowels which follow the rightmost non-high vowel, the ATR harmony applies as usual. For the portion of 1PL-FUT in (3c), the disharmony conditions are met and –ATR exponency occurs; as for the other half, the itive prefix vowel is +ATR due to the verb’s +ATR vowel.

To account for the conditioned disharmony, I let the rule –ATR-insertion (4) apply before the ATR agreement (2):

(4) –ATR-insertion

Search INR-i: [+syllabic, –high]

INR-j: [+syllabic, +high, –low];<sup>5</sup> DIR-j: left; TRM-j: [+syllabic]

Change INR-i  $\sqcup$  {–ATR}, if  $[(\exists x)(x \in \text{CINR-j})]$ <sup>6</sup>

‘Unify a non-high vowel with –ATR, if there exists a high vowel which looks leftward for a vowel but does not find it.’

The unification of –ATR with any non-high vowels is conditioned by the Search-j. By letting each high vowel look leftward for another vowel, the purpose is to check whether the string-first vowel is high or not. And the three possible scenarios are described below:

scenario i—there are no high vowels

Since there are no initiators, it is meaningless to talk about success/failure. The set CINR-j is empty, so the if-condition is not met.

scenario ii—all the high vowels are the second or any of the following vowels of the string

Each initiator of Search-j is always preceded by another vowel and successfully finds its terminator. The set CINR-j is thus empty, so the if-condition is not met.

scenario iii—the first vowel of the string is a high vowel

Due to the unsuccessful search initiated by the string-first high vowel, the set CINR-j is nonempty. Hence, the if-condition is met, so all the non-high vowels are unified with –ATR.

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<sup>5</sup> Since a high vowel is not low, this specification may need some comments. Following Hale and Reiss (2008: 96-97, 100, 274-275), I will assume that what is specified for INR and TRM is derived by means of set intersection across the segments involved. Here, the valued features shared among the high vowels are +syllabic, +high, and –low, so the natural class is displayed as such. All the rules in the current study follow this practice.

<sup>6</sup> I abuse the notation INR-j here for it to stand for the parameter of the Search-j and the set which contains the initiators which successfully find a vowel to their left. Then, the notation for complement (C) is prefixed to the set INR-j (CINR-j) to denote the set which contains the initiators which fail to find a vowel to their left.

Up to this point, with the underlying representation assumption and the rules (4) and (2) (in that order), I, U, and O are no longer present. Nonetheless,  $\text{Æ}$  has only become either A ( $\{+\text{syllabic}, -\text{high}, -\text{round}, -\text{ATR}\}$  with no specification for Low and Back) or E ( $\{+\text{syllabic}, -\text{high}, -\text{round}, +\text{ATR}\}$  with no specification for Low and Back). In order to complete the current analysis, a context-free rule is proposed, where the parameters DIR and TRM are left blank and not mentioned:

(5) exchange addition

Search INR:  $[+\text{syllabic}, -\text{high}, -\text{round}, \alpha\text{ATR}]$

Change INR  $\sqcup \{-\alpha\text{low}, -\alpha\text{back}\}$

‘Unify an ATR-specified non-high unrounded vowel with the opposite value to that of ATR for the features Low and Back.’

The application of the above rule (non-vacuously) results in a from A and e from E. And this rule should be the last to apply for the needed ATR value is cumulated from the other two: (4)  $<$  (2)  $<$  (5).

To make concrete the foregoing discussion, the derivation of  $\text{b}\text{u}-\text{b}\text{a}-\text{d}\text{i}-\text{w}\text{u}$  ‘1PL-FUT-ITIVE-climb’ from (3c) is given:

(6) an annotated derivation

underlying form:  $\text{b}\text{U}-\text{b}\text{Æ}-\text{d}\text{I}-\text{w}\text{u}$

–ATR-insertion (4)

|        |                                                                                    |          |                             |
|--------|------------------------------------------------------------------------------------|----------|-----------------------------|
| Search | INR-i: $[+\text{syllabic}, -\text{high}] - \text{Æ}$                               |          |                             |
|        | INR-j: $[+\text{syllabic}, +\text{high}, -\text{low}]$                             |          | TRM-j: $[+\text{syllabic}]$ |
|        | u                                                                                  | leftward | I                           |
|        | I                                                                                  | finds    | $\text{Æ}$                  |
|        | U                                                                                  |          | nothing                     |
| Change | the set C <sub>INR-j</sub> : {U} is a singleton, so INR-i $\sqcup \{-\text{ATR}\}$ |          |                             |
|        | $\text{Æ}$ becomes A                                                               |          |                             |

intermediate form:  $\text{b}\text{U}-\text{b}\text{A}-\text{d}\text{I}-\text{w}\text{u}$

ATR agreement (2)

|        |                           |           |                                             |
|--------|---------------------------|-----------|---------------------------------------------|
| Search | INR: $[+\text{syllabic}]$ |           | TRM: $[+\text{syllabic}, \alpha\text{ATR}]$ |
|        | U                         | rightward | A                                           |
|        | A                         | finds     | u                                           |
|        | I                         |           | u                                           |

|        | u                                                                               | nothing |
|--------|---------------------------------------------------------------------------------|---------|
| Change | INR $\sqcup$ $\{\alpha\text{ATR}\}$                                             |         |
|        | U becomes $\text{ʊ}$ (A is $-\text{ATR}$ )                                      |         |
|        | A does not change ( $-\text{ATR}$ of A and $+\text{ATR}$ of u are incompatible) |         |
|        | I becomes i (u is $+\text{ATR}$ )                                               |         |
|        | u does not change                                                               |         |

intermediate form:  $\text{bʊ-bA-di-wu}$

exchange addition (5)

|        |                                                                                         |
|--------|-----------------------------------------------------------------------------------------|
| Search | INR: $[+\text{syllabic}, -\text{high}, -\text{round}, \alpha\text{ATR}] \text{---A}$    |
| Change | INR $\sqcup$ $\{-\alpha\text{low}, -\alpha\text{back}\}$                                |
|        | A (is unified with $\{+\text{low}, +\text{back}\}$ and) becomes a (A is $-\text{ATR}$ ) |

surface form:  $\text{bʊ-ba-di-wu}$

It is common to see that multiple searches within a rule are connected (i.e. the  $x$ -ors of a search as the  $x$ -ors of another).<sup>7</sup> The above analysis, nevertheless, shows that it needs not to be the case. On the other hand, what remains constant is seeking for it is this property that enables us to detect the presence of a given configuration before a change may take place.

Appendix: inclusion of the rounding harmony

Based on McCollum and Essegbey (2020), there exists a four-way alternation  $\text{e} \sim \text{o} \sim \text{a} \sim \text{ɔ}$  (7a-d) for some prefixes in Tutrugbu, as shown in (7) for the future prefix.<sup>8</sup>

(7) rounding harmony (McCollum et al. 2020: 219; McCollum and Essegbey 2020: 5-6, in the former's transcription system)

- |    |                                         |    |                                        |
|----|-----------------------------------------|----|----------------------------------------|
| a. | $\text{e-be-fe}$ '3SG-FUT-grow'         | b. | $\text{o-bo-fe}$ '2SG-FUT-grow'        |
| c. | $\text{a-ba-ba}$ '3SG-FUT-come'         | d. | $\text{ɔ-bɔ-ba}$ '2SG-FUT-come'        |
| e. | $\text{no-ti-bo-wu}$ '2P-NEG-FUT-climb' | f. | $\text{nɔ-tɪ-bɔ-ba}$ '2P-NEG-FUT-come' |
| g. | $\text{bo-ba-ti}$ '1P-FUT-know'         |    |                                        |

<sup>7</sup> For a toy rule which neutralizes the height to mid when a vowel is preceded by a nasal and followed by a rhotic, one possible formalization of the search portion is as follows:

|        |                                                                                                            |
|--------|------------------------------------------------------------------------------------------------------------|
| Search | INR-i: $[+\text{syllabic}]$ ; DIR-i: left; TRM: $[+\text{consonantal}, +\text{sonorant}, +\text{nasal}]$   |
|        | INR-j: INR-i; DIR-j: right; TRM: $[+\text{consonantal}, +\text{sonorant}, -\text{nasal}, -\text{lateral}]$ |

If INR-j is simply set to be  $[+\text{syllabic}]$ , the vowels referred to by INR-i and those by INR-j need not be the same, which is not exactly what the toy rule requires.

<sup>8</sup> The aforementioned  $\text{e} \sim \text{a}$  alternation is subsumed under this alternation.

The ATR-distinction between  $a \sim \text{ɔ}$  and  $e \sim o$  was discussed in the earlier paragraphs. The round-distinction between  $a \sim e$  and  $\text{ɔ} \sim o$  is due to the mid rounded vowel contained within some farther preceding prefixes; in (7g), the high vowel of the first person plural prefix is rounded but does not change the non-high vowel of the future prefix into its rounded counterpart. It is shown in (7e-f) that the non-high vowels involved need not be within the two adjacent syllables. I speculate below how this alternation could be incorporated into what has been proposed.

To begin with, the underlying representation for this four-way allomorphy is assumed to be  $\{+\text{syllabic}, -\text{high}\}$ , with no specification for the features Low, Back, Round, and ATR. To account for the rounded realization, the following rule is proposed:

(8) W-coloring

Search INR:  $[+\text{syllabic}, -\text{high}]$ ; DIR: left; TRM:  $[+\text{syllabic}, -\text{high}, -\text{low}, +\text{back}, +\text{round}]$

Change INR  $\sqcup \{+\text{round}, +\text{back}, -\text{low}\}$

‘Unify +round, +back, and -low with a non-high vowel which is preceded by a mid back rounded vowel.’

Recall that Search may not succeed, so, after the application of (8), the roundedness may still remain unspecified. For this and other lack of specification, I will modify the rule (5) and change it into:

(5') exchange plus

Search INR:  $[+\text{syllabic}, -\text{high}, \alpha\text{ATR}]$

Change INR  $\sqcup \{-\alpha\text{low}, -\alpha\text{back}, -\text{round}\}$

‘Unify an ATR-specified non-high vowel with -round and the opposite value to that of ATR for the features Low and Back.’

The aim for +round to surface sometimes forces (8) to transpire before (5'), that is,  $(8) < (5')$ .

As a wrap-up, how to derive the  $o \sim \text{ɔ} \sim e \sim a$  alternation is demonstrated. Although the ascertainable rule ordering is:  $\left. \begin{matrix} (4) < (2) \\ (8) \end{matrix} \right\} < (5')$ , the demonstration, for ease of typography, begins with (8).

(9) schematic derivation of the four-way alternation

|       |                                      |
|-------|--------------------------------------|
| input | $\{+\text{syllabic}, -\text{high}\}$ |
|-------|--------------------------------------|

|                                         |                                                                                                                                                 |                                                                                                                                                 |                                                                                                                                                 |                                                                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| W-coloring (8)                          | $\left\{ \begin{array}{c} +\text{syllabic} \\ -\text{high} \\ +\text{round} \\ +\text{back} \\ -\text{low} \end{array} \right\}$                | $\left\{ \begin{array}{c} +\text{syllabic} \\ -\text{high} \\ +\text{round} \\ +\text{back} \\ -\text{low} \end{array} \right\}$                | <i>search failure<br/>and no change</i>                                                                                                         |                                                                                                                                                 |
| –ATR-insertion (4)<br>ATR agreement (2) | $\left\{ \begin{array}{c} +\text{syllabic} \\ -\text{high} \\ +\text{round} \\ +\text{back} \\ -\text{low} \\ +\text{ATR} \end{array} \right\}$ | $\left\{ \begin{array}{c} +\text{syllabic} \\ -\text{high} \\ +\text{round} \\ +\text{back} \\ -\text{low} \\ -\text{ATR} \end{array} \right\}$ | $\left\{ \begin{array}{c} +\text{syllabic} \\ -\text{high} \\ +\text{ATR} \end{array} \right\}$                                                 | $\left\{ \begin{array}{c} +\text{syllabic} \\ -\text{high} \\ -\text{ATR} \end{array} \right\}$                                                 |
| exchange plus (5')                      | <i>unification failure and/<br/>or vacuous unification</i>                                                                                      |                                                                                                                                                 | $\left\{ \begin{array}{c} +\text{syllabic} \\ -\text{high} \\ +\text{ATR} \\ -\text{round} \\ -\text{low} \\ -\text{back} \end{array} \right\}$ | $\left\{ \begin{array}{c} +\text{syllabic} \\ -\text{high} \\ -\text{ATR} \\ -\text{round} \\ +\text{low} \\ +\text{back} \end{array} \right\}$ |
| output                                  | o                                                                                                                                               | ɔ                                                                                                                                               | e                                                                                                                                               | a                                                                                                                                               |

In the row pertinent to ATR, no differentiation is made for the sources of –ATR: due to the string-first vowel being high (4) or because of a stem’s first vowel being –ATR (2). The source of +ATR, nonetheless, can only be attributed to the +ATR of the stem-first vowel (2).

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